

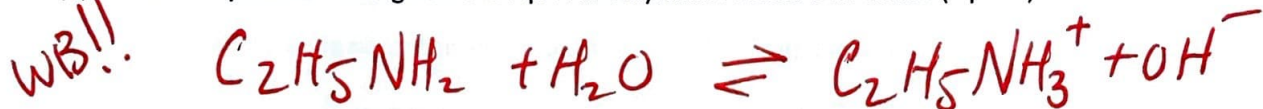
Quiz Free Response Practice (2009B, #1, 11 points)

A pure 14.85 g sample of the weak base ethylamine, $C_2H_5NH_2$, is dissolved in enough distilled water to make 500. mL of solution.

- (a) Calculate the molar concentration of the $C_2H_5NH_2$ in the solution. (2 points)

$$M = \frac{n}{V} = \frac{0.3293}{0.500} = \underline{\underline{0.659M}} \quad n = \frac{14.85g}{45.10 \frac{g}{mol}} = 0.3293 \text{ mol}$$

- (b) Write an equation showing how the aqueous ethylamine reacts with water. (1 point)



- (c) Write the equilibrium-constant expression for the reaction between $C_2H_5NH_2(aq)$ and water. (1 point)

$$K_b = \frac{[C_2H_5NH_3^+][OH^-]}{[C_2H_5NH_2]}$$

- (d) Of $C_2H_5NH_2(aq)$ and $C_2H_5NH_3^+(aq)$, which is present in the solution at the higher concentration at equilibrium? Justify your answer. (1 point)

~~R $C_2H_5NH_2$~~ ~~Because~~

The weak base $C_2H_5NH_2$ is present at a higher concentration b/c the K_b for the base is very small, $K_b \ll 1$, and very little of the base dissociates w/ H_2O to form $C_2H_5NH_3^+$.

- (e) A different solution is made by mixing 500. mL of 0.500 M $C_2H_5NH_2$ with 500. mL of 0.200 M HCl. Assume that volumes are additive. The pH of the resulting solution is found to be 10.93.

- a. Calculate the concentration of $OH^-(aq)$ in the solution. (1 point)

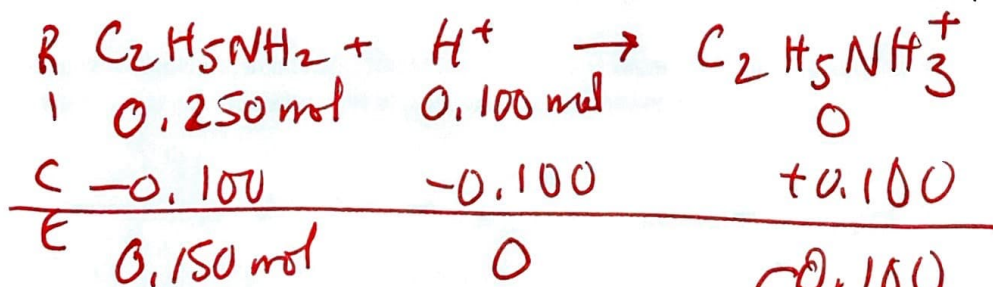
$$pOH = 14 - pH = 14 - 10.93 = 3.07$$

$$[OH^-] = 10^{-pOH} = \underline{8.51 \times 10^{-4} M}$$

- b. Write the net-ionic equation that represents the reaction that occurs when the $C_2H_5NH_2$ solution is mixed with the HCl solution. (1 point)



- c. Calculate the molar concentration of the $C_2H_5NH_3^+(aq)$ that is formed in the reaction. (2 points)



$$[C_2H_5NH_3^+] = \frac{\text{mol}}{\text{Vol}} = \frac{0.100 \text{ mol}}{(0.500 L + 0.500 L)} = 0.100 M$$

- d. Calculate the value of K_b for $C_2H_5NH_2$. (2 points)

$$K_b = \frac{[C_2H_5NH_3^+][OH^-]}{[C_2H_5NH_2]}$$

$[C_2H_5NH_2] = \frac{0.150 \text{ mol}}{(0.500 + 0.500)} = 0.150 M$

$$= \frac{0.100 \cdot 8.5 \times 10^{-4}}{0.150} = \underline{5.7 \times 10^{-4}}$$

(H.H.): $pOH = pK_b + \log \frac{[HA]}{[A^-]}$

$$3.07 = pK_b + \log \left(\frac{0.100}{0.150} \right)$$

$$3.07 = pK_b + (-0.176)$$

$$pK_b = 3.246$$

$$K_b = 10^{-pK_b} = 10^{-3.246} = 5.7 \times 10^{-4}$$

AB1

1. Nitrogen monoxide, NO(g) , can undergo reactions to produce acids such as HNO_2 , a weak acid with a K_a of 4.0×10^{-4} and a $\text{p}K_a$ of 3.40.

- a. A student is asked to make a buffer solution with a pH of 3.40 by using 0.100 M $\text{HNO}_2(\text{aq})$ and 0.100 M $\text{NaOH}(\text{aq})$.

- ii. Explain why the addition of 0.100 M $\text{NaOH}(\text{aq})$ to 0.100 M $\text{HNO}_2(\text{aq})$ can result in the formation of a buffer solution. Include the net ionic equation for the reaction that occurs when the student adds the $\text{NaOH}(\text{aq})$ to the $\text{HNO}_2(\text{aq})$. (2 points)

Strong base changes weak acid to CB

NaOH will neutralize the HNO_2 to become the CB, NO_2^-

$$\text{OH}^- + \text{HNO}_2 \rightarrow \text{H}_2\text{O} + \text{NO}_2^-$$

← CB of HNO_2

- ii. Determine the volume, in mL, of 0.100 M $\text{NaOH}(\text{aq})$ the student should add to 100. mL of 0.100 M $\text{HNO}_2(\text{aq})$ to make a buffer solution with a pH of 3.40. Justify your answer. (2 points)

$\text{p}K_a = 3.40$
 $\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$

b/c $\text{pH} = \text{p}K_a$, $[\text{HA}] = [\text{A}^-]$.
HALF of the $[\text{HA}]$ must change to A^-

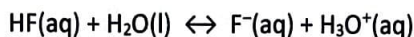
50 mL ~~NaOH~~ 0.100 M NaOH
 $= 0.005 \text{ mol } \text{NaOH}$,
 reacting $\frac{1}{2}$ of the $0.100 \cdot 0.100 = 0.010 \text{ mol } \text{HNO}_2$

- b. A second student makes a buffer by dissolving 0.100 mol of $\text{NaNO}_2(\text{s})$ in 100. mL of 1.00 M $\text{HNO}_2(\text{aq})$. Which is more resistant to changes in pH when a strong acid or a strong base is added, the buffer made by the second student or the buffer made by the first student in part (a)? Justify your answer. (1 point)

*2nd Student $[\text{HNO}_2] = \underline{1.00 \text{ M}}$ $[\text{NO}_2^-] = \frac{0.100}{0.100} = \underline{1.00 \text{ M}}$
 Greater buffer capacity than the 1st student (higher concentration)

1st student:
 $[\text{NO}_2^-] = \frac{0.005 \text{ mol } \text{NaNO}_2}{0.150 \text{ L}} = 0.033 \text{ M}$ $[\text{HNO}_2] = \frac{0.005 \text{ mol}}{0.150} = 0.033 \text{ M}$

#B2



2. The ionization of HF(aq) in water is represented by the equation above. In a 0.0350 M HF(aq) solution, the percent ionization of HF is 13.0 percent.

- a. Two particulate representations of the ionization of HF molecules in the 0.0350 M HF(aq) solution are shown below in Figure 1 and Figure 2. Water molecules are not shown. Explain why the representation of ionization of HF molecules in water in Figure 1 is more accurate than the representation in Figure 2. (1 point)

(The key below identifies the particles in the representations.)

Fig 1 shows the majority of HF as undissociated, which is accurate for weak acids like HF

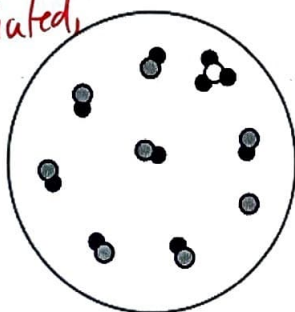


Figure 1

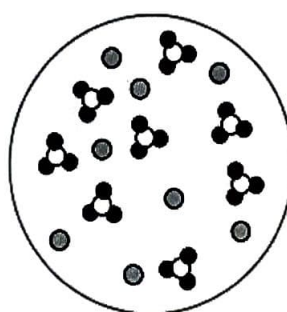


Figure 2

Fig 2 shows ALL of the HF as dissociated which would only be true if it were a strong acid.

- b. Use the percent ionization data above to calculate the value of K_a for HF. (2 points)

$$\% \text{ Ion} = \frac{[\text{H}^+]_{\text{eq}}}{[\text{HF}]} \times 100 = 13\%$$

$$\frac{[\text{H}^+]}{0.0350} = 0.13$$

$$[\text{H}^+] = 0.00455$$

$$K_a = \frac{[\text{H}^+][\text{F}^-]}{[\text{HF}]}$$

$$= \frac{(0.00455)^2}{(0.0350 - 0.00455)} = 6.8 \times 10^{-4}$$

- c. If 50.0 mL of distilled water is added to 50.0 mL of 0.0350 M HF(aq), will the percent ionization of HF(aq) in the solution increase, decrease, or remain the same? Justify your answer with an explanation or a calculation. (1 point)

%I increases for lower concentrations.



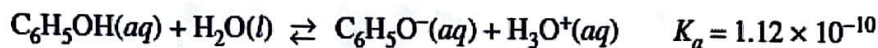
$$K_a = 6.8 \times 10^{-4}$$

$$Q = \frac{[\text{H}^+] \times 0.5 \times [\text{F}^-] \times 0.5}{[\text{HF}] \times 0.5}$$

$$Q = 0.5 \times K$$

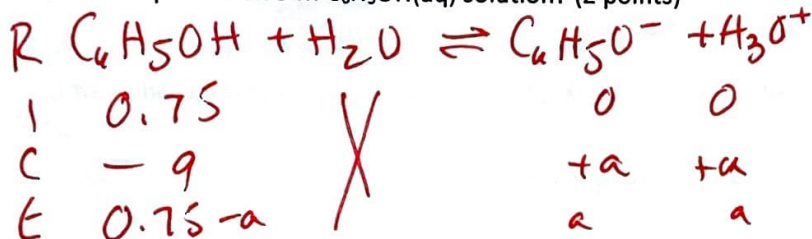
$Q < K$, rxn shifts FORWARD, $[\text{H}_3\text{O}^+]$ increases

AB3



3. Phenol is a weak acid that partially dissociates in water according to the equation above.

a. What is the pH of a 0.75 M $\text{C}_6\text{H}_5\text{OH}(\text{aq})$ solution? (2 points)



$$K_a = \frac{[\text{H}_3\text{O}^+][\text{C}_6\text{H}_5\text{O}^-]}{[\text{C}_6\text{H}_5\text{OH}]}$$

$$1.12 \times 10^{-10} = \frac{a^2}{(0.75-a)} \quad \text{a is small!}$$

$$a = 9.165 \times 10^{-6} = [\text{H}_3\text{O}^+]$$

$$\text{pH} = -\log(9.165 \times 10^{-6}) = \underline{\underline{5.04}}$$

b. For a certain reaction involving $\text{C}_6\text{H}_5\text{OH}(\text{aq})$ to proceed at a significant rate, the phenol must be primarily in its deprotonated form, $\text{C}_6\text{H}_5\text{O}^-(\text{aq})$. In order to ensure that the $\text{C}_6\text{H}_5\text{OH}(\text{aq})$ is deprotonated, the reaction must be conducted in a buffered solution. On the number scale below, circle each pH for which more than 50 percent of the phenol molecules are in the deprotonated form, $\text{C}_6\text{H}_5\text{O}^-(\text{aq})$. Justify your answer. (2 points)

1 2 3 4 5 6 7 8 9 10 11 12 13 14

$$\text{p}K_a = -\log 1.12 \times 10^{-10} = 9.95$$

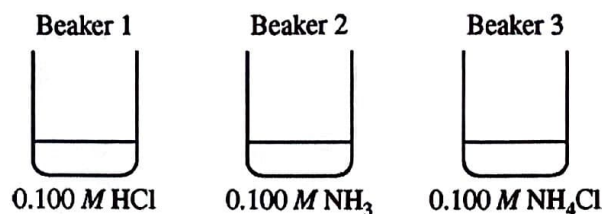
$$\text{pH} = \text{p}K_a = 9.95 \quad \text{when } [\text{HA}] = [\text{A}^-]$$

$$\text{pH} < 9.95 \quad [\text{HA}] > [\text{A}^-]$$

$$\text{pH} > 9.95 \quad [\text{HA}] < [\text{A}^-]$$

more than 50% A^- !

AB5



2. Each of three beakers contains 25.0 mL of a 0.100 M solution of HCl, NH₃, or NH₄Cl, as shown above. Each solution is at 25°C.

- a. Determine the pH of the solution in beaker 1. Justify your answer. (1 point)

$$\text{pH} = -\log [\text{H}^+] = -\log 0.100 = 1.000 \quad (\text{H}^+ = \text{HCl} \text{ b/c HCl completely dissociates} \rightarrow \text{strong acid})$$

- b. In beaker 2, the reaction $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \leftrightarrow \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$ occurs. The value of K_b for $\text{NH}_3(\text{aq})$ is 1.8×10^{-5} at 25°C.

- i. Write the K_b expression for the reaction of $\text{NH}_3(\text{aq})$ with $\text{H}_2\text{O}(\text{l})$. (1 point) $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$

$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$

- ii. Calculate the $[\text{OH}^-]$ in the solution in beaker 2. (2 points)

$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} = 1.8 \times 10^{-5}$$

a is wee!

$$\frac{a^2}{(0.100 - a)} = 1.8 \times 10^{-5}$$

$$a = 0.00134 \text{ M} = [\text{OH}^-]$$

c. In beaker 3, the reaction $\text{NH}_4^+(\text{aq}) + \text{H}_2\text{O}(\text{l}) \leftrightarrow \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$ occurs.

i. Calculate the value of K_a for $\text{NH}_4^+(\text{aq})$ at 25°C . (1 point)

$$K_a = K_w / K_b = 1 \times 10^{-14} / 1.8 \times 10^{-5} = \underline{5.6 \times 10^{-10}}$$

ii. The contents of beaker 2 are poured into beaker 3 and the resulting solution is stirred. Assume that volumes are additive. Calculate the pH of the resulting solution. (2 points)

$$[\text{NH}_3] = [\text{NH}_4^+] = \text{BUFFER!!}$$

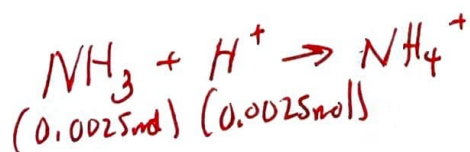
$$\text{pOH} = \text{p}K_b = -\log 1.8 \times 10^{-5} = 4.74$$

$$\text{pH} = 14 - \text{pOH} = \underline{\underline{9.26}}$$

d. The contents of beaker 1 are poured into the solution made in part (c)(ii). The resulting solution is stirred. Assume that volumes are additive. Is the resulting solution an effective buffer? Justify your answer. (1 point)

$$25 \text{ mL} \cdot 0.100 \text{ mol/L } \text{NH}_3 = 0.00250 \text{ mol } \text{NH}_3$$

$$25 \text{ mL} \cdot 0.100 \text{ mol/L } \text{H}^+ = 0.00250 \text{ mol } \text{H}^+$$



No. All of the weak base has been converted to the conjugate NH_4^+ , so there remains, only NH_4^+ present.

NOT a BUFFER